Sketch2Style:
A Controllable AI Fashion Design Using Diffusion Models & Sketch Conditioning

Group 3:

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Date: 7th May 2026

Course: AI688-001_BK_Spring 2026

Professor: Reda Nacif El Alaoui



LONG ISLAND UNIVERSITY

Source: https://cdn.outrank.so/badd1fc9-54db-49f3-a872-8c2c738b8342/1e8f8422-6230-4366-b98e-41c542222dbf/what-shoes-to-wear-with-midi-dresses-dress-sneakers.jpg

OUTLINES

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1. Introduction
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4. Literature Review Results
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8. Understanding the Functions
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INTRODUCTION

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- Fashion design begins with conceptual sketches
- Designers require both structural and stylistic control
- Diffusion models excel at photorealistic image generation
- However, they lack precise spatial control
- **Our solution**: Sketch-conditioned diffusion for spatially controlled fashion generation



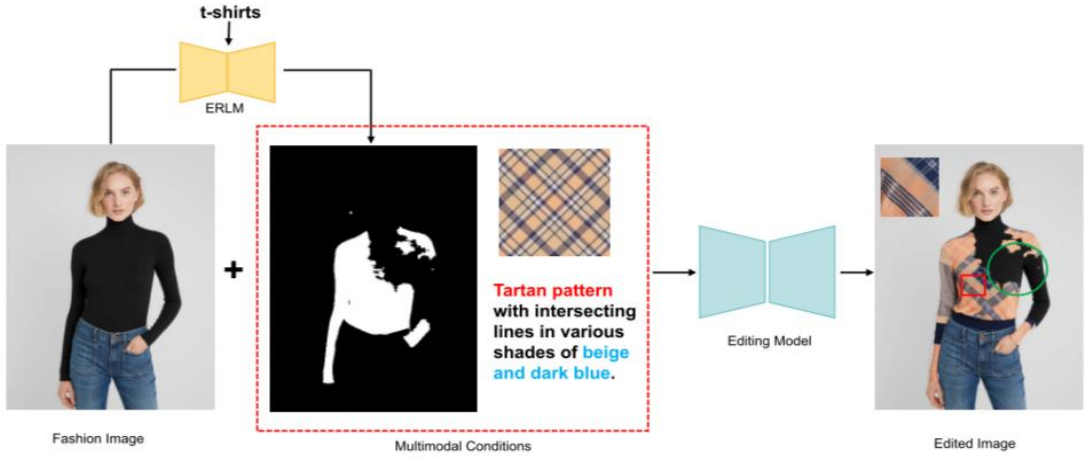
Source: https://davincinext.com/2024/01/22/ai-web3-fashion-revolution-202/

Problem Statement

Current AI tools suffer from:

- ❌ Cannot preserve sketch geometry
- ❌ Outputs are misaligned with designer intent

Source:
https://www.themoonlight.io/en/review/dpdedit-detail-preserved-diffusion-models-for-multimodal-fashion-image-editing



Our solution directly addresses each gap:

- ✓ Sketch-conditioned generation using ControlNet
- ✓ Full model fine-tuning for domain adaptation
- ✓ Gradio interface for real-time interaction

APPLICATION

- **AI-assisted fashion prototyping** – Accelerate design iterations
- **Rapid design exploration** – Generate multiple variations quickly
- **Cost reduction** – Minimize physical sample production
- **Support for small designers** – Democratize access to AI tools
- **Creative exploration tool** – Expand design possibilities

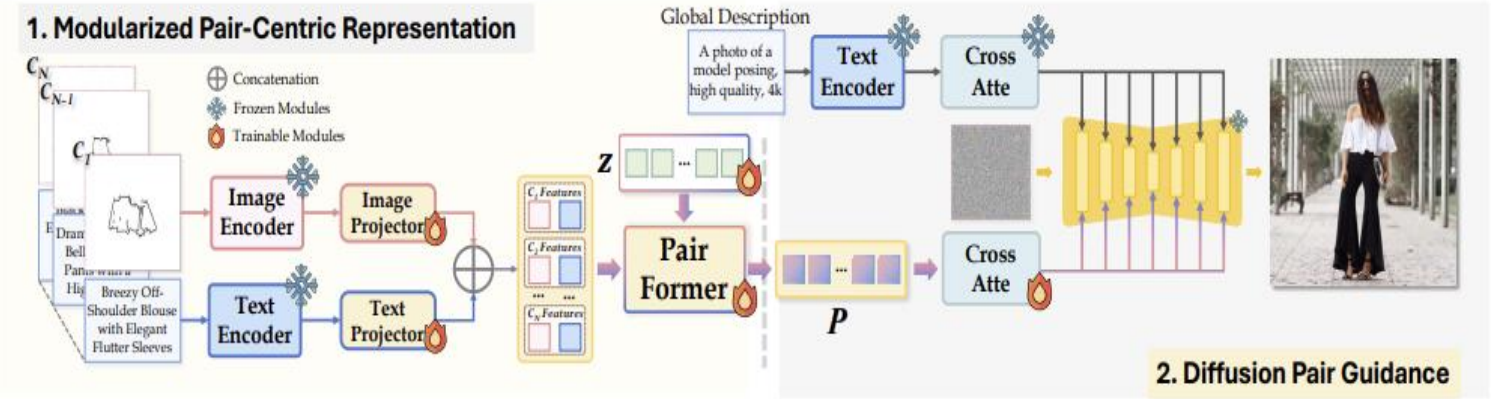


Source: https://www.mirrorsize.com/blogs/custom-fits-in-clicks-ais-role-in-made-to-measure-apparel

LITERATURE REVIEW

Paper 1: LOTS (ICCV 2025)

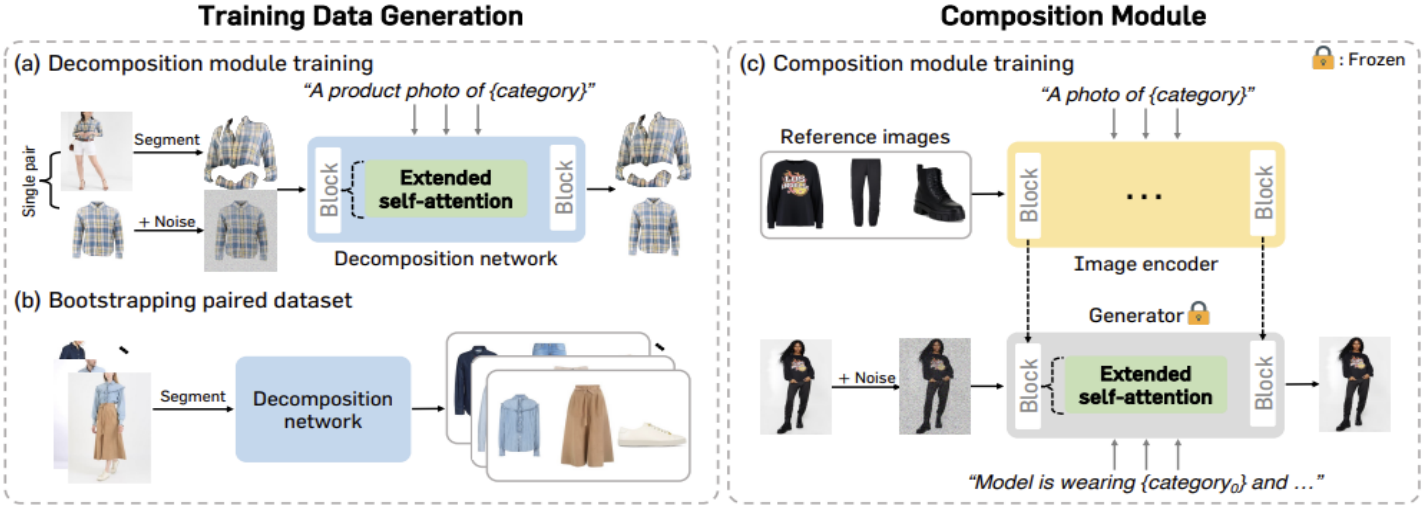
- **Title:** LOTS of Fashion
- Multi-conditioning diffusion
- Sketch + Text pairing
- Modular attention-based merging
- Introduced Sketchy dataset
- Improved attribute localization



Source: F. Girella et al.

Paper 2: BootComp (CVPR 2025)

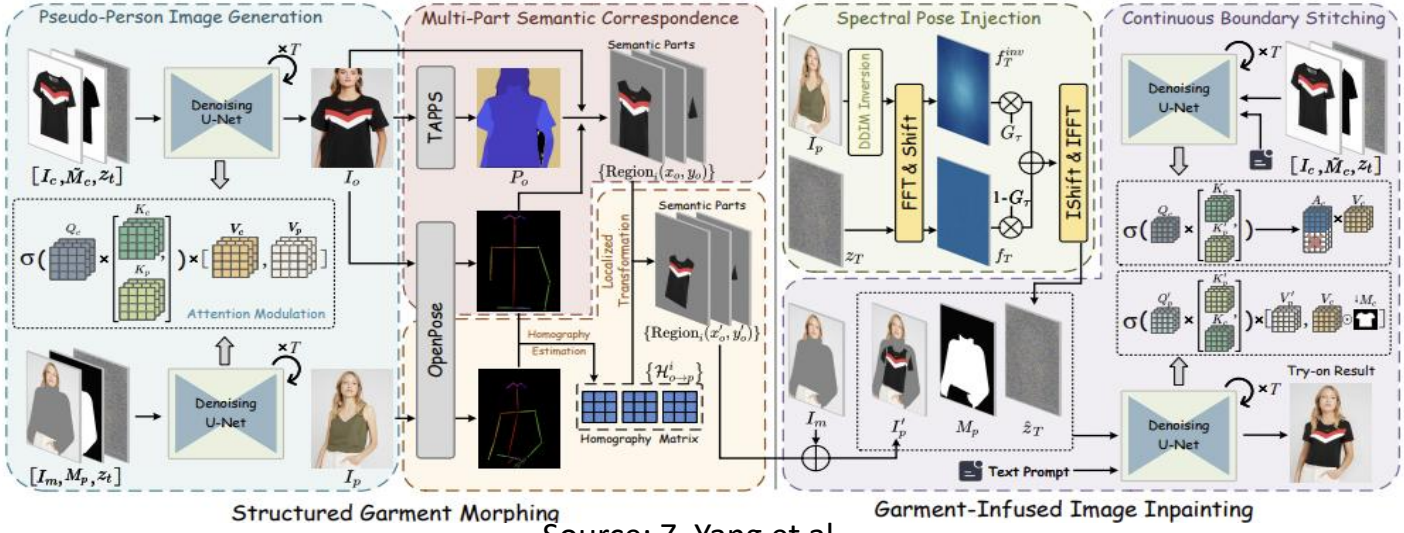
- **Title:** Controllable Human Image Generation
- Multi-garment conditioning
- Two-stage pipeline
- Synthetic paired data generation
- Extended self-attention
- +30% improvement in MP-LPIPS(Multi-Scale Learned Perceptual Image Patch Similarity)



Source: Y. Choi et al.

Paper 3: OmniVTON (ICCV 2025)

- **Title:** Training-Free Universal Virtual Try-On
- Structured Garment Morphing
- Spectral Pose Injection
- Continuous Boundary Stitching
- Training-free approach
- Strong pose consistency



Source: Z. Yang et al.

LITERATURE REVIEW RESULTS

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Aspect	LOTS	BootComp	OmniVTON
Objective	Sketch-text generation	Multi-garment control	Training-free try-on
Conditioning	Sketch + Text	Multi-garment images	Pose + garment
Training	Fully trained	Trained	Training-free
Strength	Layout accuracy	Texture fidelity	Pose consistency
Limitation	Complex design	Not sketch-driven	Not design-focused

Research Gap

- No ControlNet-based fashion sketch system
- No simple **interactive sketch-based interface**
- No simple designer-friendly implementation
- No system with **full diffusion fine-tuning**

Our contribution: First ControlNet-based fashion generation on Fashion-MNIST with accessible web interface

ALGORITHM/APPROACH/METHODOLOGY

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Step 1 Dataset Exploration	Fashion-MNIST: 70,000 images, 10 clothing categories, Balanced dataset Split: 9,600 training, 2,400 validation, 2,000 testing
Step 2 Data Preprocessing	Resize 28×28 → 256×256, Convert grayscale → RGB, Normalize: [1,-1], Apply Canny edge detection, Generate sketch maps
Step 3 Text Prompt Construction	Convert labels to descriptive prompts Encode using CLIP Provide semantic control
Step 4 Model Pipeline	Edge map → ControlNet, Prompt → CLIP encoder, Random noise → Stable Diffusion v1.5, Dual conditioning: spatial (sketch) + semantic (text), VAE decoder → Final image
Step 5 Training Strategy	Each sample includes: Target image, Edge map, Text prompt Training process: Add noise to latent, Predict noise residual, Train ControlNet + UNet + Text Encoder + VAE layers

OVERALL WORKFLOW

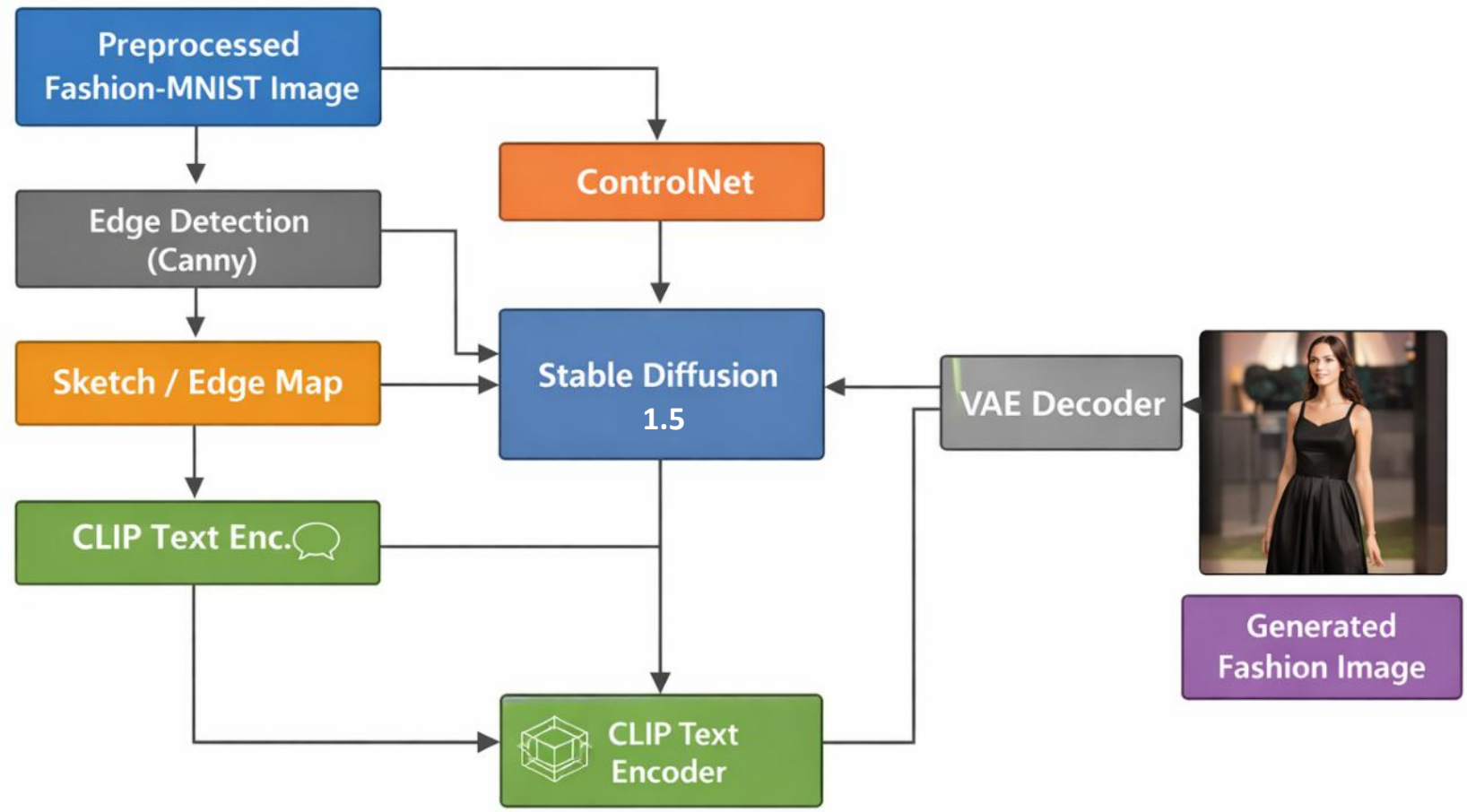


Figure 1: Overall Workflow. All components are fully trainable with differentiated learning rates.

MODEL ARCHITECTURE

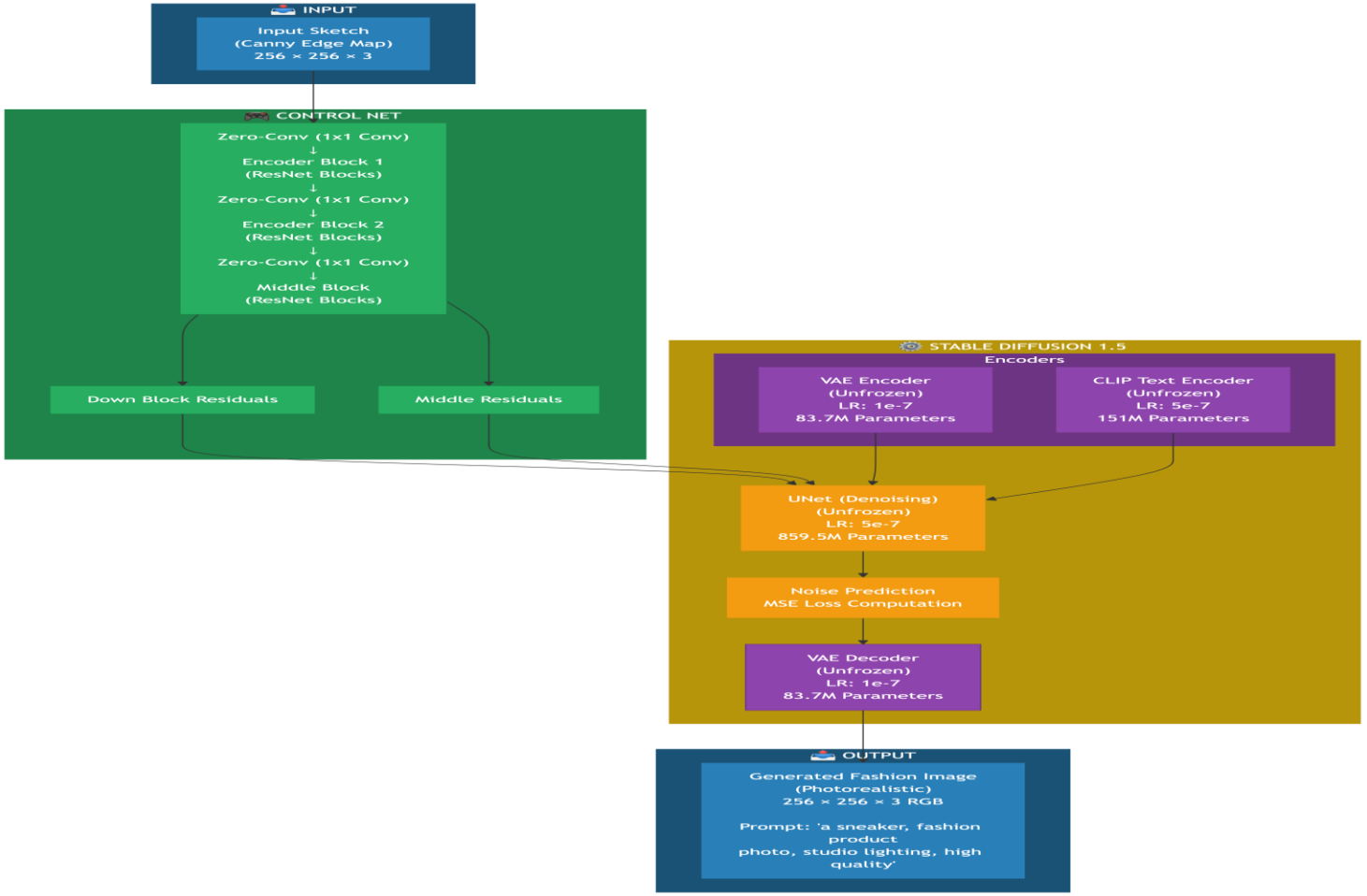


Figure 2: Sketch2Style Model Architecture with Full Model Unfreezing

DATASET STATISTICS

Dataset Name: Fashion-MNIST Overview

- Total Images: 70,000
- Image Type: 28 x 28 grayscale → affects realism
- Categories: 10 balanced classes (7,000 each)
- Pixel Range: 0-255
- Final training: 9,600 samples for 14,400 steps.

Selected Categories : Dress & Sneaker

DATASET DISTRIBUTION (SELECTED CATEGORIES)

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Dataset Split After Filtering

Type	Samples
Training	9,600
Validation	2,400
Test	2,000
Total	**16,000**

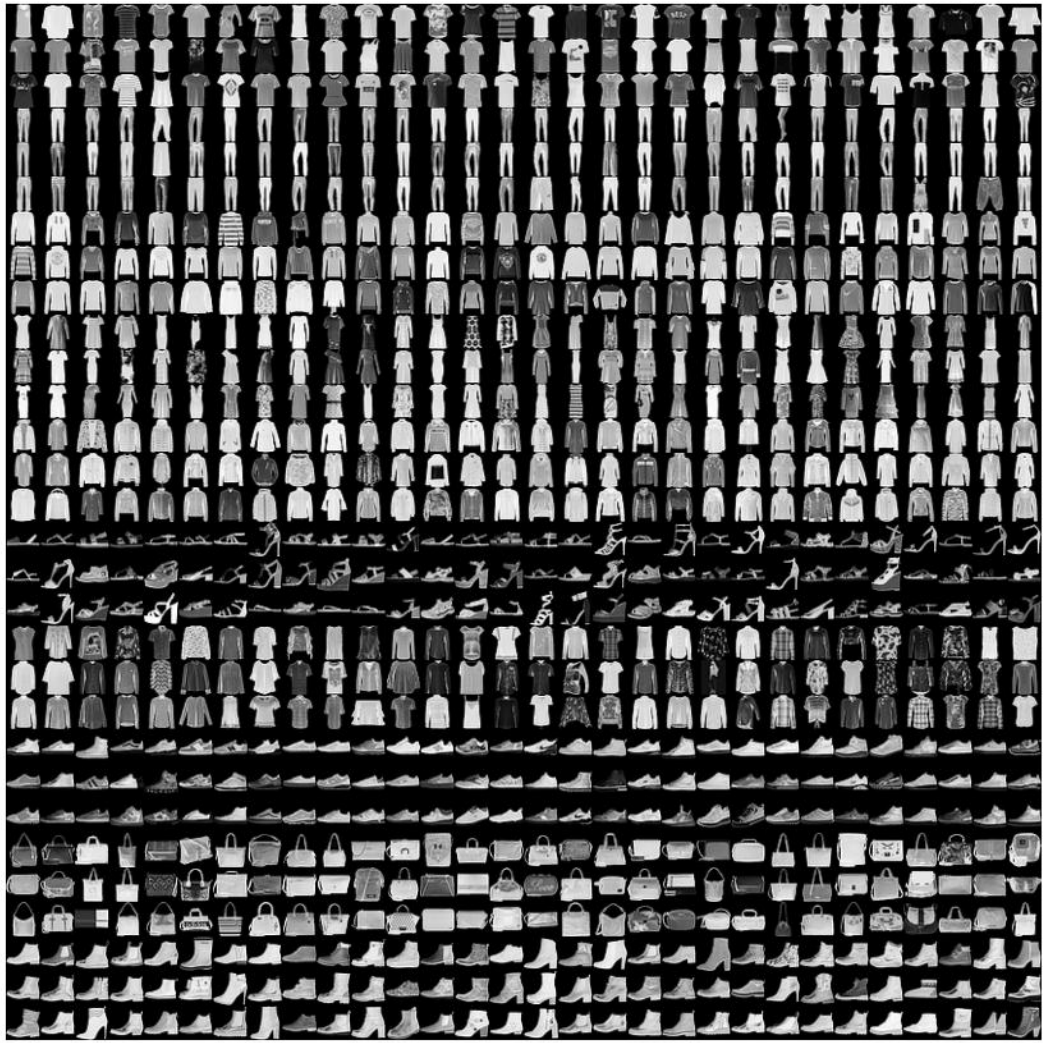
Why 5,600 instead of 7,000?

- Original Fashion-MNIST has 7,000 images per category
- 20% reserved for validation (1,400 per category)
- Remaining 80% used for training (5,600 per category)

Per Category Breakdown

Category	Training	Validation	Test	Total
Dress	5,600	1,400	1,000	8,000
Sneaker	5,600	1,400	1,000	8,000
Combined	**11,200**	**2,800**	**2,000**	**16,000**

DATASET EXPLORATION



Source: https://www.kaggle.com/code/gpreda/cnn-with-tensorflow-keras-for-fashion-mnist

PRE-TRAINED MODEL SELECTIONS

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Stable Diffusion Versions

Version	Text Encoder	Resolution	Parameters	Key Characteristics
v1.4	CLIP ViT-L/14	512×512	860M	First widely released version
v1.5	CLIP ViT-L/14	512×512	860M	Improved image quality, best ControlNet compatibility
v2.0	OpenCLIP	768×768	865M	Larger text encoder, higher resolution
v2.1	OpenCLIP	768×768	865M	Refined version of v2.0

Selected: v1.5 because it offers best compatibility with ControlNet and lower computational requirements for T4 GPU training.

ControlNet Variants

Variant	Conditioning Input	Best Use Case	Selected
Canny	Edge maps	Sketch preservation, structural control	YES
Depth	Depth maps	3D structure preservation	NO
Pose	Human keypoints	Human pose control	NO
Scribble	Rough sketches	Freehand drawing	NO
MLSD	Line detection	Architectural drawings	NO
Normal	Normal maps	Surface detail control	NO

Selected: Canny variant because Fashion-MNIST edge maps closely match designer sketches and provide precise structural guidance for garment shapes.

Model Parameter Summary (100% Trainable)

Component	Parameters	Trainable	Status
Stable Diffusion UNet	859,520,964	859,520,964	Trainable
Stable Diffusion VAE	83,653,863	83,653,863	Trainable
CLIP Text Encoder	123,060,480	123,060,480	Trainable
ControlNet	361,279,120	361,279,120	Trainable
TOTAL	**1,427,514,427**	**1,427,514,427**	**100.00% Trainable**

Key Insight: Final implementation uses full model unfreezing with 100% trainable parameters.

UNDERSTANDING THE FUNCTIONS

Role of Each Model

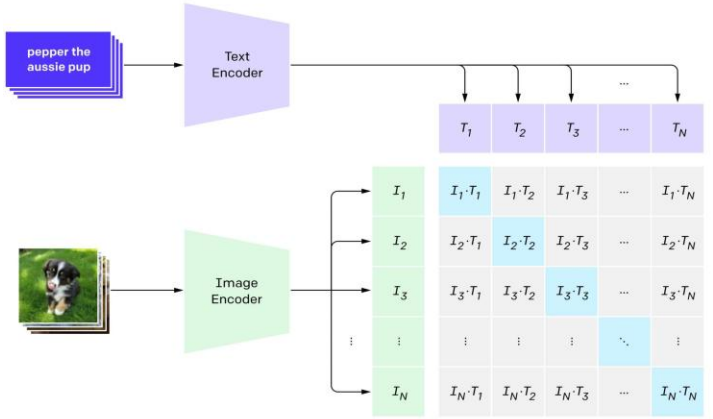
- **Stable Diffusion v1.5:**
  Generates realistic images with UNet



Source: https://medium.com/the-bytedoodle-blog/canny-edge-detection-deb0d62a8003

- **ControlNet:**
  Enforces spatial structure

- **CLIP(Contrastive Language-Image Pre-training):**
  Guides semantic meaning by text prompts
  Used Constant prompt:
  Ex: "a sneaker, fashion product photo, studio lighting, high quality"



- **VAE (Variational Autoencoder)**:
  Encodes and decodes images

Source: https://viso.ai/deep-learning/clip-machine-learning/

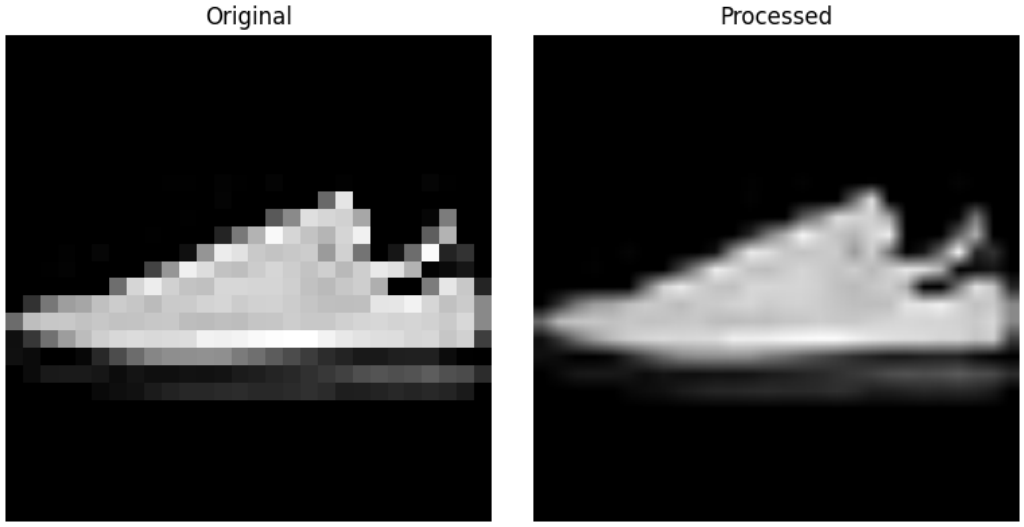
TRAINING STRATEGY

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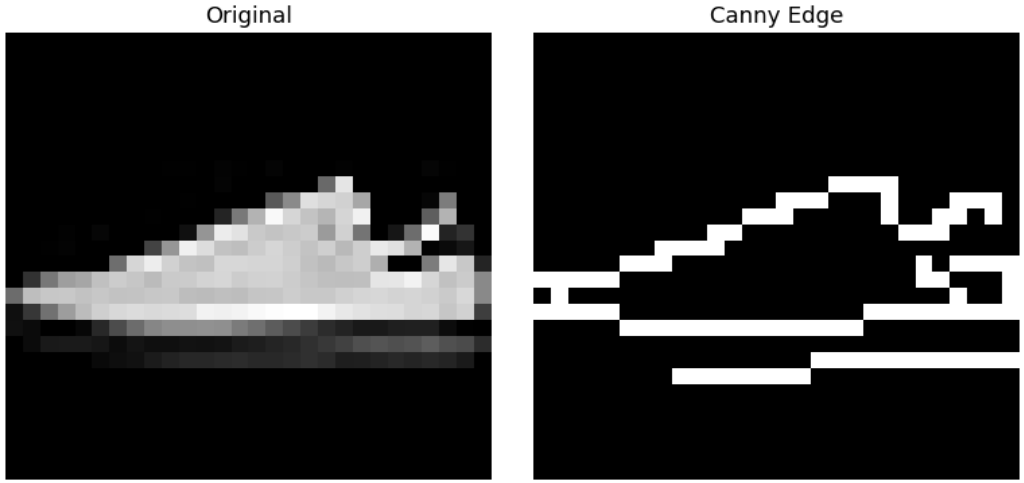
- Full model fine-tuning (100% trainable)
- **Learning rates:**
  - ControlNet → 1e-6
  - UNet/Text Encoder → 5e-7
  - VAE → 1e-7
- **Batch size** = 2
- **Total Steps** = 14,400 (for 3 epochs)
- **Total Parameters** = 1.43B
- Stable convergence with no NaN or gradient issues

OBTAINED RESULTS (PREPROCESSING)

Preprocessing



Canny Edge Detection



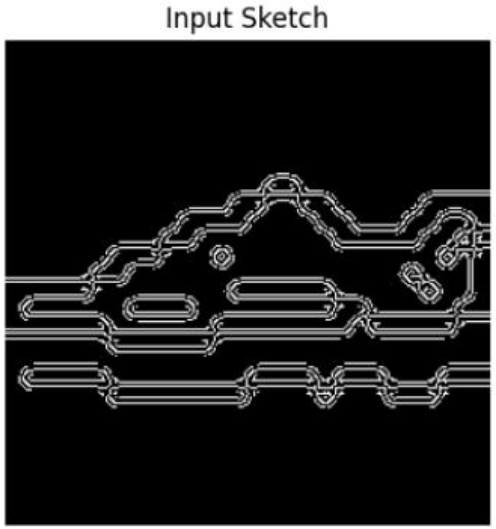
OBTAINED RESULTS (TRAINING/TESTING)

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Training

```
Starting Epoch 3/3
Epoch 3/3 | Batch 200/4800 | Global Step 9800 | Loss: 0.0475
Epoch 3/3 | Batch 400/4800 | Global Step 10000 | Loss: 0.0002
Epoch 3/3 | Batch 600/4800 | Global Step 10200 | Loss: 0.0020
Epoch 3/3 | Batch 800/4800 | Global Step 10400 | Loss: 0.0135
Epoch 3/3 | Batch 1000/4800 | Global Step 10600 | Loss: 0.0111
Epoch 3/3 | Batch 1200/4800 | Global Step 10800 | Loss: 0.0025
Epoch 3/3 | Batch 1400/4800 | Global Step 11000 | Loss: 0.0084
Epoch 3/3 | Batch 1600/4800 | Global Step 11200 | Loss: 0.0031
Epoch 3/3 | Batch 1800/4800 | Global Step 11400 | Loss: 0.0217
Epoch 3/3 | Batch 2000/4800 | Global Step 11600 | Loss: 0.0007
Checkpoint saved: checkpoints/full_epoch_3_step_2000.pt
Epoch 3/3 | Batch 2200/4800 | Global Step 11800 | Loss: 0.0015
Epoch 3/3 | Batch 2400/4800 | Global Step 12000 | Loss: 0.0038
Epoch 3/3 | Batch 2600/4800 | Global Step 12200 | Loss: 0.0069
Epoch 3/3 | Batch 2800/4800 | Global Step 12400 | Loss: 0.0079
Epoch 3/3 | Batch 3000/4800 | Global Step 12600 | Loss: 0.0022
Epoch 3/3 | Batch 3200/4800 | Global Step 12800 | Loss: 0.0003
Epoch 3/3 | Batch 3400/4800 | Global Step 13000 | Loss: 0.0037
Epoch 3/3 | Batch 3600/4800 | Global Step 13200 | Loss: 0.0046
Epoch 3/3 | Batch 3800/4800 | Global Step 13400 | Loss: 0.0031
Epoch 3/3 | Batch 4000/4800 | Global Step 13600 | Loss: 0.0080
Checkpoint saved: checkpoints/full_epoch_3_step_4000.pt
Epoch 3/3 | Batch 4200/4800 | Global Step 13800 | Loss: 0.0016
```

Image Generation or Testing

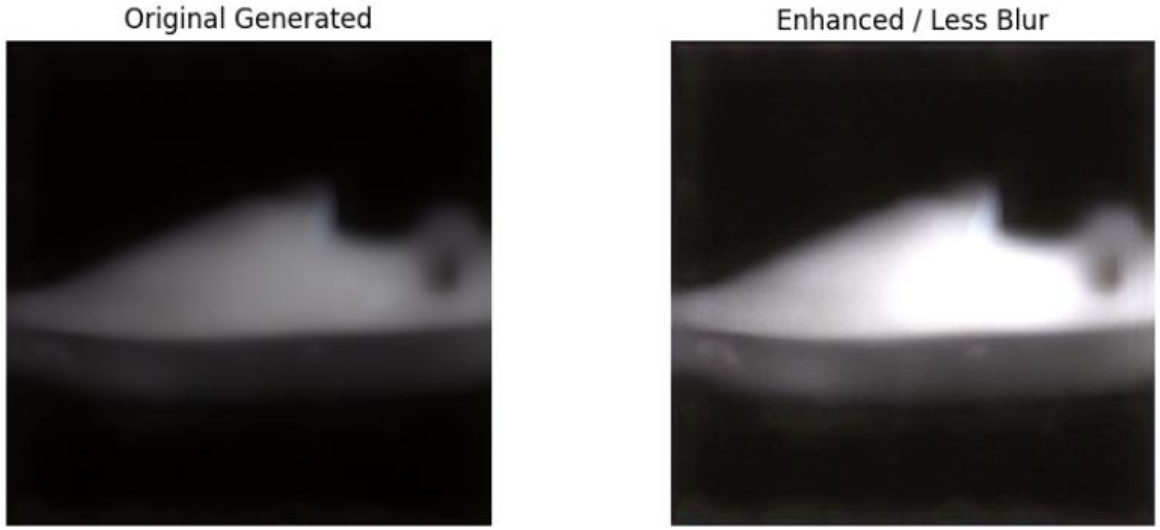




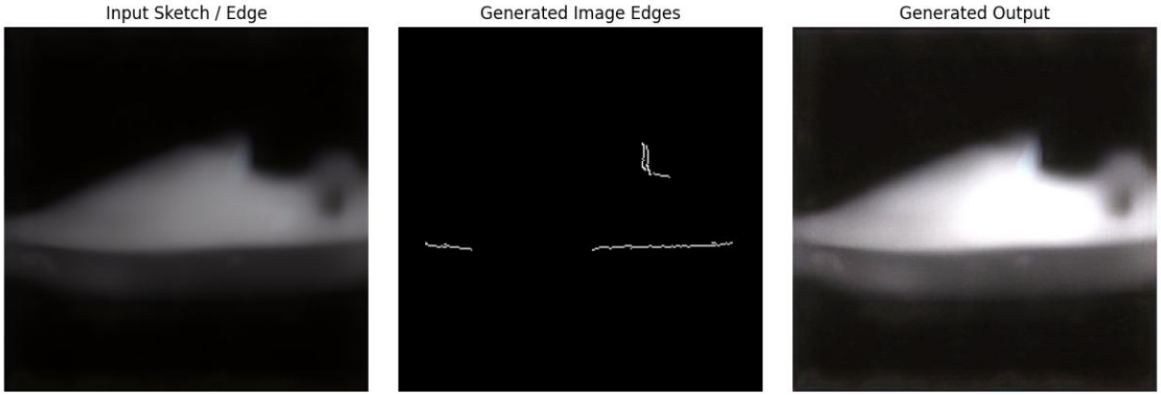
OBTAINED RESULTS (SHARPEN IMAGE)

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Sharpen Image



Evaluation



Laplacian sharpening improves perceived clarity without altering structural constraints.

OBTAINED RESULTS (CLEAN VS MESSY SKETCH)

Comparison b/w Clean Sketch & Messy Sketch

Clean Sketch (User Input)



Output from Clean Sketch



Messy Sketch (User Input)



Output from Messy Sketch



Model demonstrates robustness to degraded sketches while maintaining overall silhouette.

OBTAINED RESULTS (STABLE DIFFUSION VS OUR MODEL)

Comparison b/w Stable Diffusion & Our Model



Stable Diffusion generates realistic images but ignores sketch structure. Our **Sketch2Style** model preserves exact edge constraints through **ControlNet**, proving that spatial conditioning is essential for design tasks.

OBTAINED RESULTS (COMPARISON TABLE - SD VS OURS)

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Comparison of text-only Stable Diffusion versus our sketch-conditioned Sketch2Style model

Aspect	Stable Diffusion (Text-Only)	Ours (Sketch2Style)
Input Conditioning	Text prompt only	Text prompt + Canny edge map
Focuses on	Realistic, ignores sketch	Follows sketch, less detailed
Architecture	SD v1.5 UNet + VAE + Text Encoder	SD v1.5 + ControlNet (full unfreeze)
User Interface	None	Gradio web interface
Robustness Testing	Not tested	Noise + blur + dilation

Key advantage: Explicit spatial control via sketch conditioning

OBTAINED RESULTS (TRAINING RESULT + METRICS TABLE)

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➢ Training Result:

- Initial loss ≈ 0.0475
- Final loss ≈ 0.0132
- Minimum Loss ≈ 0.0002
- No NaN issues
- Stable convergence

➢ Observations:

- Preserves garment structure
- Generates realistic shapes
- Text prompt influence visible
- Model remains stable
- Recognizable output
- Slight Quality Drop

Metric	Value
SSIM (Structural Similarity Index Measure)	0.2217
CLIP Score (Contrastive Language-Image Pre-training)	0.2021
FID (Frechet Inception Distance)	253.21
Inference Time	2-3 sec

- ❑ **SSIM** → structure preserved
- ❑ **CLIP** → moderate semantic alignment
- ❑ **FID** → high due to dataset mismatch
- ❑ Model shows stable convergence without NaN or gradient instability.

OBTAINED RESULTS (CLIP/FID LIMITATIONS)

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- **CLIP** limited due to:
  - low resolution
  - grayscale dataset

- **FID** high because:
  - 28×28 vs 256×256 mismatch
  - RGB vs grayscale

These limitations highlight dataset constraints rather than model failure.

COMPUTATIONAL RESOURCES

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Hardware:

- GPU: NVIDIA RTX PRO 6000 Blackwell Server Edition
- VRAM: ~48 GB
- Platform: Google Colab

Training Configuration:

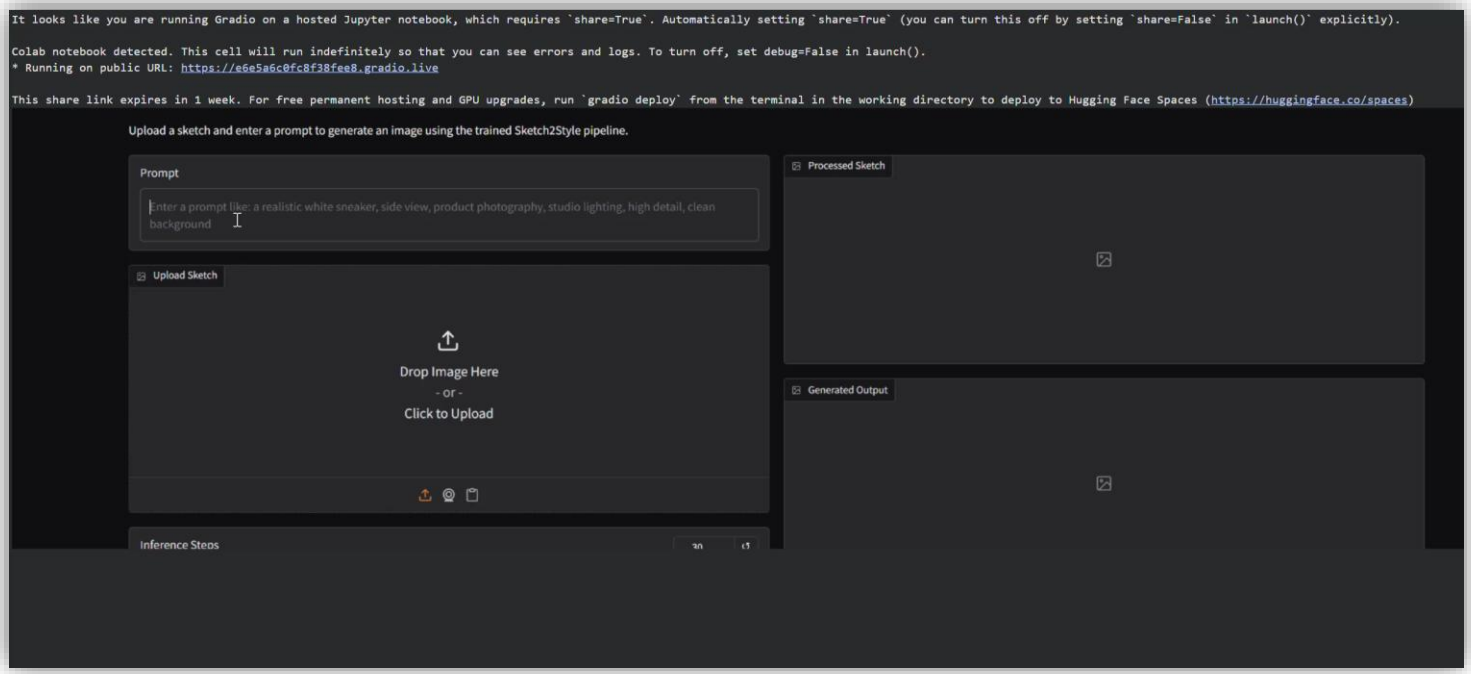
- Training Time: ~3 hours
- Total Steps: 14,400 (3 epochs)
- Batch Size: 2 (training), 1 (inference)
- Precision: float32

Memory Usage:

- Training: ~14 GB VRAM
- Inference: ~8.5 GB VRAM Environmental Impact:
- Carbon Footprint: ~2.5 kg CO2
- Equivalent to: Driving 10 miles in a passenger vehicle

OBTAINED RESULTS (GRADIO INTERFACE)

➢ Gradio Interface



Features:

- Real-time prompt input
- Sketch upload via drag-and-drop
- Interactive generation with adjustable parameters
- Public shareable link (expires in 1 week)

Users can upload sketches and enter prompts to generate fashion images instantly.

COMPARISON WITH BASE PAPER (LOTS OF FASHION)

Aspect	LOTS of Fashion [8] (Base Paper)	Ours (Sketch2Style)
Base Model	Stable Diffusion v1.5	Stable Diffusion v1.5
Conditioning	Sketch-Text Pairs	Canny Edge Maps + Text
Fine-tuning	Full model	Full model (100% unfrozen)
Learning Rates	Single LR	Differentiated LRs (1e-6, 5e-7, 1e-7)
Dataset	Fashion datasets	Fashion-MNIST
Categories	Multiple	Dress + Sneaker
CLIP Score	~0.30	0.2021
Inference Time	~4 seconds	2-3 seconds
Interactive UI	No	Yes (Gradio)
Robustness Testing	No	Yes
Open Source	Partial	Complete

CONCLUSION & FUTURE WORK

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Conclusion:

- Successfully built sketch-conditioned system
- ControlNet enables structural control
- Trade-off: Increased control leads to slightly reduced photorealism
- Model prioritizes structural control over photorealistic detail
- Demonstrates effectiveness of ControlNet for spatially constrained generation

Future Work:

- Increase resolution to 512×512
- Use DeepFashion dataset for color and texture learning
- Improve CLIP Score to > 0.30
- Reduce FID to < 100



Source: https://www.lezhougarment.com/ultimate-guide-to-3d-fashion-design-the-future-of-fashion/

ETHICAL CONSIDERATIONS

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Aspect	Summary
Data Ethics	Public Fashion-MNIST dataset; no human subjects; 70,000 images across 10 categories
Model Ethics	Open source (MIT License); reproducible with fixed random seeds; balanced representation
Responsible Use	Assists designers (not replaces); human-in-the-loop control; no autonomous decisions; designs require human review before production
Scientific Integrity	Results reported honestly; all prior work properly cited
Dual Use	Academic research only; assists designers, not replaces creativity
Environmental Impact	2.5 kg CO_2 (equivalent to driving 10 miles)

TEAM-WORK DISTRIBUTION

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Team Member	Primary Contributions	Responsibilities
RUPALI	• Project Lead & Architecture Design • Literature Review Coordination • Model Architecture Design • Final Report Compilation	• Led weekly team meetings and coordinated project timeline • Design ControlNet + Stable Diffusion architecture • Coordinate literature review across 10 papers • Integrate all sections into final proposal
KARTAVYA	• Data Acquisition & Preprocessing • Edge Detection Pipeline • Implementation Lead • Results Documentation	• Download and process Fashion-MNIST dataset • Implement Canny edge detection pipeline • Develop Gradio web interface • Document technical implementation details

TIMELINE

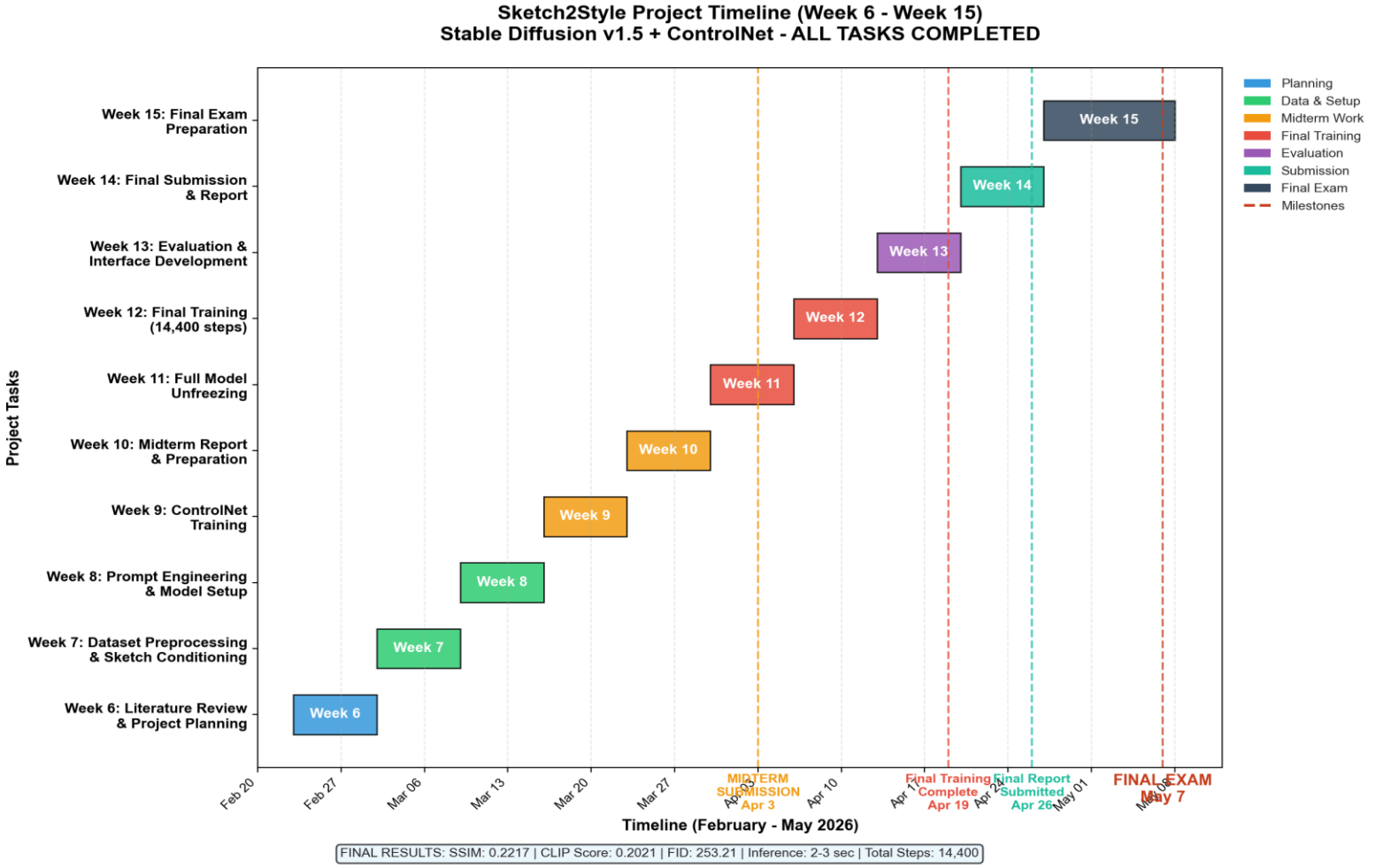


Figure 3: Gantt Chart

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[2] L. Zhang et al., "Adding Conditional Control to Text-to-Image Diffusion Models," ICCV, 2023.

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[7] Y. Choi et al., "Controllable Human Image Generation with Personalized Multi-Garments," CVPR, 2025.

[8] Z. Yang et al., "OmniVTON: Training-Free Universal Virtual Try-On," ICCV, 2025.

Thank You ..!
Any Questions ?